Which Inputs Carry a Style? Custom Style Transfer and Its Cost in Real-Time Diffusion on Consumer Hardware

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Abstract

This case study on real-time img2img style transfer asks which properties an input image must retain for the trained style to appear in the output. I trained a rank-16 LoRA on 150 renders of my 3D artwork and ran it in StreamDiffusionTD on an 8 GB laptop GPU. Eight inputs, from the Cycles render to greyscale noise, were streamed with and without the adapter and rated on four fixed criteria. The adapter improved every condition, but similarity to the training data did not predict by how much. A flat colour render costing no render time scored two points below a path-traced render at a minute per frame. With all 150 training captions identical, the trigger token never separated from the subject. The adapter cost no measurable speed or memory: 7.5–8.0 fps either way. One denoising step raised this to 11.8 fps; generating the input on the same GPU cost 15 %.

1. Introduction

1.1. Motivation

The motivation for this project originates from an artwork I created as branding for a music festival. Produced in Blender and rendered with the Cycles path-tracing engine, it consists of abstract organic shapes generated through procedural sculpting. The main form alone carries about 916,000 triangles, and one 1920×1080 frame took about 30 seconds at 1000 samples, roughly twelve hours for a minute of animation. Long sequences were therefore impractical, and live use was difficult.

This limitation motivated me to test an alternative workflow: training a custom LoRA adapter on the artwork's visual style and driving a real-time img2img pipeline with a simplified, faster-to-render representation of the original artwork. The adapter could then be used to reintroduce stylistic properties such as material appearance and lighting during inference.

It is unclear how much visual information an input must retain for the learned style to appear. In a real-time img2img pipeline, the input constrains the output structure, while the style is contributed by the adapter, so simplifying the input may either weaken style expression or leave more room for it to emerge. This question also extends to inputs that no longer depict the original scene, as in my case, where the inputs consist of noise or abstract structures generated in TouchDesigner. In addition, personalisation may introduce computational costs that matter on consumer hardware.

Training on my own renders is also a deliberate position. Instead of prompting a general model towards a look I had in mind, I wanted a system shaped by work I made myself, so that the aesthetic decisions stay mine and the model carries them out. Grimes and Harrison describe this stance as treating the model as an assistant working under the artist's direction rather than as an autonomous generator [GH26].

This case study is therefore guided by the following question:

Research Question. Which properties of the input image determine whether a custom-trained style appears in the output of a real-time diffusion pipeline, and what does personalisation cost on consumer hardware?

Contributions. (i) An eight-condition comparison of input representations, each run with and without a self-trained style adapter. (ii) The finding that neither similarity to the training data nor the input contrast I measured predicts how strongly the style appears. What the strongest conditions appear to share is a figure clearly separated from its ground. (iii) A cost profile showing that the adapter is not the expense, while the number of denoising steps is the lever on throughput. The limit is not memory but sharing the GPU with the image source. (iv) The negative result that identical training captions left the trigger token tied to the subject, so the prompt could not switch the style on or off.

2. Related Work

Running diffusion at interactive rates rests on a stack of accelerations, all of them used here. Denoising happens in a compressed latent space, consistency distillation reduces the number of steps required, and LCM-LoRA delivers that acceleration as an inference-time adapter [RBL*22, LTH*23, LTP*23]. StreamDiffusion assembles these into a streaming pipeline, and its TouchDesigner integration provides the deployment context of this study [KXH*23].

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What such a pipeline produces depends not only on the model but on the image it is given. The input is encoded into the latent space and partially replaced with noise before denoising, and the noise level bounds how far the result can depart from it [MHS*22]. It is less clear how far the input decides whether a trained style comes through at all. Akten et al. report that darker input regions lead their system to draw on darker learned features and brighter regions on brighter ones, so a general property of the input selects which specific learned features appear [AFG19]. They describe this effect rather than measure it, in a conditional GAN driven by a camera feed. The same may hold in a few-step diffusion pipeline with a self-trained adapter. The choice of input would then determine which parts of the trained style appear, and not only the composition of the output. It remains unclear which properties of the input account for this.

In general, Low-Rank Adaptation trains a small number of additional parameters instead of updating the full model [HSW*22], which brings artist-specific adaptation within reach of consumer hardware. Artist-led work treats a small, self-curated corpus not as a data shortage but as a deliberate source of creative influence, since changing a few examples changes how the model behaves [VPF22].

Grimes and Harrison take this furthest. Their BLK-Assist framework fine-tunes diffusion models on a single artist's own corpus and frames the result as an assistant working under the artist's direction, so that authorship stays with the artist [GH26]. Their pipeline is text-to-image and runs on a multi-GPU workstation, so neither the input nor the hardware constrains it there.

This study takes the same artist-led position into a different setting, where the structure comes from an image rather than a prompt, the hardware is a single consumer GPU, and the outputs are rated rather than only described. To the best of my knowledge, this combination has received little systematic evaluation, so what follows is a measurement of it on one style and one machine.

3. System Overview and Setup

The study comprises four recorded phases, in which eight input conditions were streamed through the pipeline described below, with and without the style adapter.

3.1. Pipeline

All runs used Stable Diffusion 1.5 ("v1-5-pruned-emaonly-fp16") inside StreamDiffusionTD 0.3.1 [Hin25], built on StreamDiffusion [cum24], running in TouchDesigner 2025.33230. Following the combination Luo et al. describe [LTP*23], two adapters were merged at startup, LCM-LoRA (slot 0, weight 1.0) for few-step sampling and a custom style adapter (slot 1, weight 0.9). Setting the first slot to weight zero also removed the effect of the second, so slot order is part of the configuration. The pipeline ran at 512×512 with TAESD, acceleration "none", "cfg_type: self", guidance scale 1.0, and the stochastic similarity filter disabled (no frame skipping), since identical still frames would otherwise be skipped and the throughput figures would measure skipping rather than generation. Hardware: Windows 11 laptop, NVIDIA RTX 5070 Laptop GPU (8 GB, Blackwell), 64 GB RAM. TensorRT, NVIDIA's toolkit for accelerating inference, was not used in this study; production support for this GPU generation is listed only from a later component release [Hin25]. ControlNet, IP Adapter and StreamV2V all require it in v0.3.1 and were therefore outside the scope of these runs.



Figure 1: *Twelve of the 150 Cycles renders used to train the style adapter.*

3.2. Style Adapter Training

The adapter was trained on 150 frames at 512×512 from the source artwork, rendered in Blender with Cycles (Fig. 1). Camera angle, distance and framing were varied across the set. All 150 images shared the caption "orgflw, abstract organic shapes, blue sky background", which was also used as the prompt across all four phases. Training used kohya_ss [bma25]; the configuration is listed in Table 1. A LoRA binds what stays constant and factors out what varies. Here the shape carries part of the style yet should not be reproduced, so the training images cover five distinct forms rather than one.

Parameter	Value	Parameter	Value
Dataset	150 renders	Optimizer	AdamW
Resolution	512×512	Schedule	cosine
Trigger token	*orgflw*	Batch / epochs / steps	1 / 10 / 3000
Rank / alpha	16 / 8	Training time	18 min
UNet LR	1×10^{-4}	Peak VRAM	4069 MiB
Text-encoder LR	5×10^{-5}	Checkpoint	epoch 9

Table 1: *Style adapter training configuration. VRAM peak occurred during sample generation. Training steps used 3106–3227 MiB.*

Since training loss plateaued from epoch 2 onward, checkpoint selection relied on visual comparison across epochs 2, 5, 7, and 9 (criteria in Section 3.5). Epochs 7 and 9 scored identically; epoch 9 was chosen for a closer material match too subtle for the five-point scale. Epoch 9 was used unchanged for all later runs (Fig. 2).

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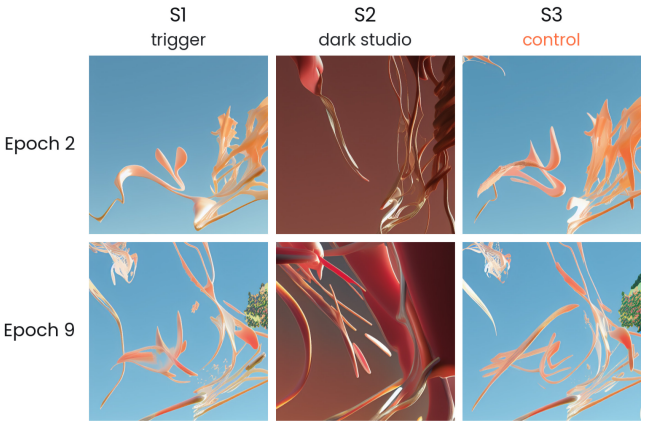


Figure 2: *Sample outputs at epochs 2 and 9 (seed 42, 20 steps, euler_a, guidance scale 7). Epochs 5 and 7 were also compared, not shown.* S1*: training caption with trigger token.* S2*: same, with a dark studio background replacing the sky.* S3*: control without the trigger token.*

3.3. Input Conditions

Eight input conditions were arranged as a cascade of decreasing similarity to the training data: the original Cycles render (C01), low-poly version (C02), Eevee render (C03), Blender viewport solid shading (C04), flat colour variant (C05), binary black-and-white mask (C06), an audio-reactive TouchDesigner structure following a tutorial by Tschepe [Tsc21] (C07), and greyscale noise (C08). All images were fed as still frames, so each condition presented the same input in every run (Fig. 3). The conditions differ in several properties at once, because isolating one would have required manual image editing, which is not a practical alternative to rendering. The ordering follows my judgement of how far each condition visually departs from the training data, and I expected style fidelity to fall along it.

3.4. Measurement Protocol

In Phase 1, all eight conditions were streamed with the style adapter active. Phase 2 repeated them without the adapter, isolating its contribution. Phase 3 reran C01 with the style adapter but only the first denoising step, isolating the second step's contribution. Phase 4 added one supplementary run in which the original network of input C07 was generated live rather than supplied as an exported still frame.

All other settings were fixed: seed 804281, prompt "orgflw, abstract organic shapes, blue sky background", step schedule "t_index_list [15, 45]", LCM-LoRA at 1.0, style adapter at 0.9, and resolution. The host machine remained plugged in, with identical open applications, untouched during runs except for timing and nvidia-smi readings.

Throughput was read from the component's server log. GPU memory, clock speed and temperature were logged via nvidia-smi. Each measurement window began 60 seconds after stream start, excluding warm-up. Memory is reported as total device allocation, an upper bound for pipeline consumption.

3.5. Rating the Outputs

Each run produced one output image, except for the supplementary run in Phase 4, where one frame was selected from the running stream. I rated the selected frames on five criteria (1–5 points each): organic morphology, colour palette and highlights, surface detail, overall stylistic coherence, and input preservation. The first four form the Style Index, with a possible range of 4–20. Input preservation is kept separate because it measures how much of the input's own structure survives, rather than how strongly the trained style comes through. The highest score of 5 points required the following characteristics in each category. Organic morphology: continuous, separated strands with the curvature of the training renders; colour palette and highlights: the salmon and pale pink material with subsurface scattering, iridescent colour shifting and the green glitch structure; surface detail: the small mesh fragments from the renders and an uneven surface; overall stylistic coherence: the image reading as one of my renders; input preservation: the input layout still recognisable. To limit rater bias, the criteria were fixed before any output was examined closely. In addition, I rated all outputs myself in a single session, in randomised order, from neutrally named files. Only clear differences are interpreted as meaningful, since the scale has no defined unit and a one-point gap may reflect nothing more than where I set the boundary between two neighbouring grades.

4. Evaluation

4.1. Style Expression Across the Cascade

Contrary to the assumption stated in Section 3.3, these scores do not follow the original cascade order. Reducing the polygon count of the source geometry had no visible effect; C01 and C02 scored identically on all four criteria. Two inputs that carry neither shading nor depth scored high. A binary mask (C06) scored 13, two points above the fully shaded Eevee render (C03) at 11, while flat colour (C05) at 12 also outscored viewport solid (C04) at 8. The conditions come out in the same order without the adapter, where the highest and lowest score are only three points apart. That range is too small to interpret on its own, but it does show that the departure from the cascade comes from the input, not the adapter (Fig. 4).

Two observations recur across the per-criterion scores. Conditions with a clearly separated figure, C01, C02, C05 and C06, occupy the upper half of the ranking, while those whose figure blends into its surroundings (C03, C04) or which spread detail evenly across the frame (C07, C08) occupy the lower half. C06 sits at the extreme of this pattern. Its figure-ground separation is the strongest in the series, and it is the only condition to reach 4 on both organic morphology and surface detail, the highest values observed. However, its colour palette score of 2 is the lowest among the well-performing conditions because its black background, the darkest in the series, remained black in the output (Fig. 3). The four conditions carrying colour scored 3 or 4 on colour palette, the four without scored 2.

Figure-ground separation here is a matter of how the image is

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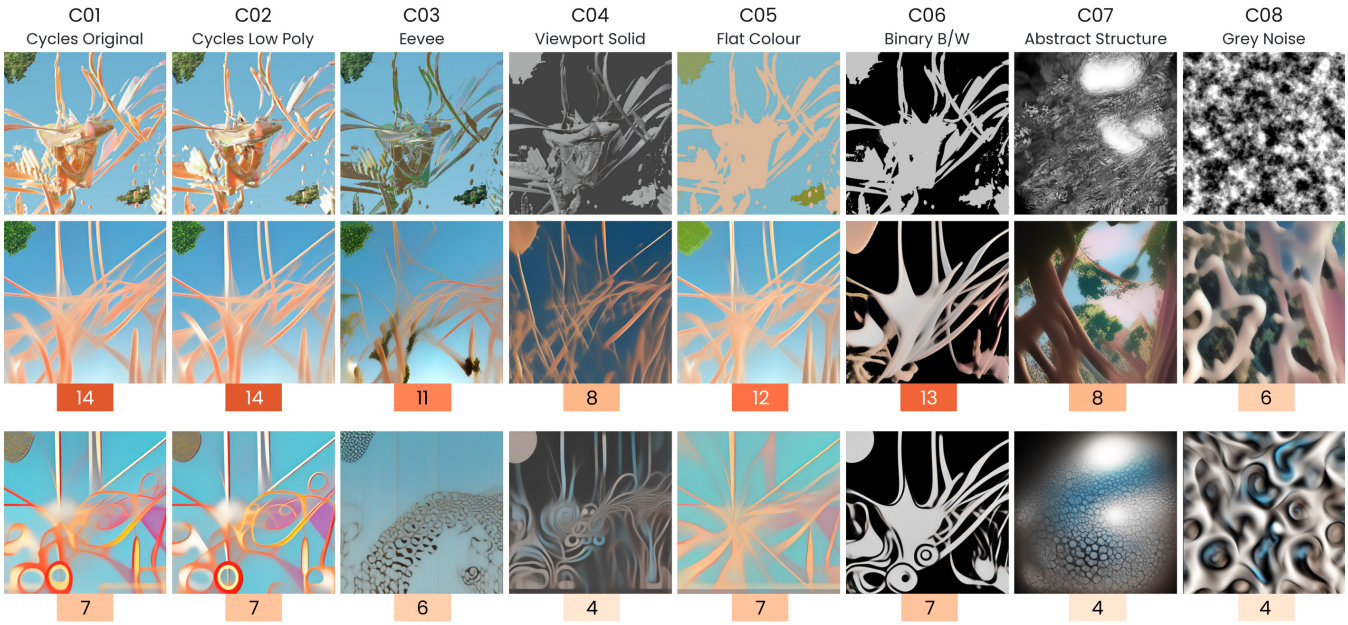


Figure 3: *The eight input conditions.* Top row*: input.* Middle row*: Phase 1–output with the style adapter.* Bottom row*: Phase 2–output without it. Each badge gives the Style Index of that run, range 4–20.*

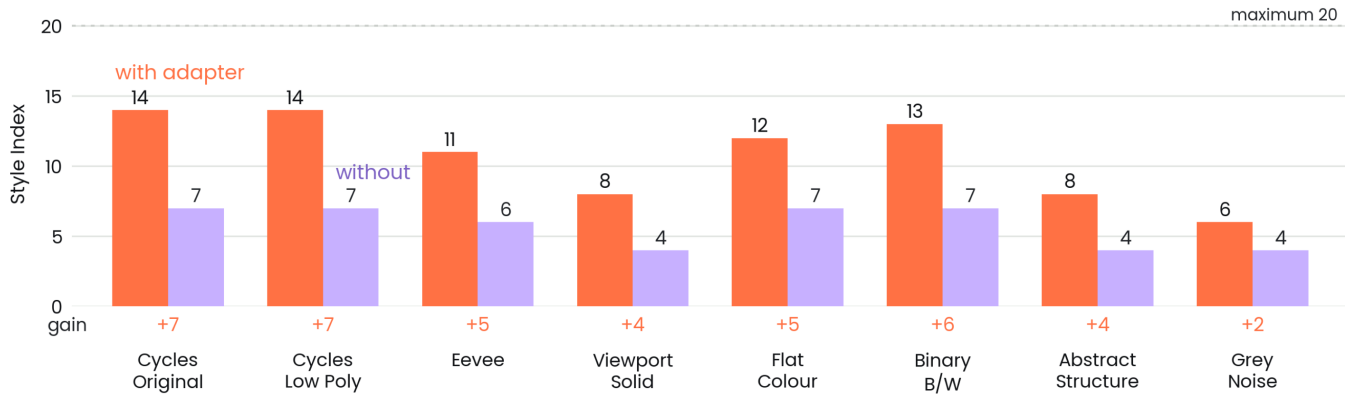


Figure 4: *Style Index for each input condition, with and without the style adapter. Sum of four criteria rated 1–5, range 4–20. The orange figure below each pair is the gain.*

organised, not of how much contrast it carries. Measured as the standard deviation of input luminance at 512×512, normalised to 0–1, the flat colour input has the lowest contrast in the series at 0.06 and the binary mask the highest at 0.38, yet both rank near the top, at 12 and 13 points on the Style Index. The greyscale noise, second highest at 0.27, scores lowest of all. Contrast therefore does not order the conditions, but a figure that stands out from its background seems to do so.

The observations below are based on a visual side-by-side comparison of the eight styled outputs. I report patterns that recur across conditions, but why they occur remains my hypothesis, untested here.

Where the input carried colour alongside a form close to the original render (C01, C02, C03, C05), both materials learned from the training data appeared in the positions they occupy in the renders, the organic forms in salmon and the frame edges as green glitch structure. Where the form stayed close but colour was absent (C04, C06), only the salmon to pale pink material appeared, carrying the form alone.

C07 and C08 carried neither colour nor the original form. Their outputs showed a wider range of colour than C04 and C06, including green structure. The absence of green structure in C04 and C06 can therefore not be attributed to the absence of colour alone. Within C07 and C08, the brightness of each region appeared to de-

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cide which elements from the training images were placed there, dark ones in dark regions and lighter ones in bright regions. My hypothesis is that large uniform areas of near-black and near-white give the adapter little to respond to, which would explain why the flat regions of C06 and C07 carry the least style, structure and colour.

C03 and C04 score lower than their place in the cascade predicts. Among the four conditions carrying depth information, they reach 2 on organic morphology against 3 for C01 and C02. I cannot say which input property accounts for this, since several differ at once.

4.2. What the Adapter Contributes

The adapter raised the Style Index in every condition, from +7 for the two Cycles conditions down to +2 for greyscale noise (Fig. 4). The conditions that gained most are also the ones that scored highest with the adapter. This may mean the adapter amplifies what the input already carries, but it remains untested here.

Input preservation stays the same or improves in six of the eight conditions and drops by one and two points in the binary mask and the audio-reactive structure, differences too small on a five-point scale to interpret.

The binary mask condition further demonstrates that the trained colours are mostly introduced by the adapter rather than inherited from the input. Although the input is strictly black and white, some colour appears in the styled output. In the matched run without the adapter, no colour was produced (Fig. 3). C07 and C08 produce colour without the adapter, but not the trained palette. Since the prompt names a blue sky background, the blue in those outputs is most plausibly prompt-driven.

Surface detail remains low throughout, reaching 4 only for the binary mask and otherwise sitting at 2 or 3. The small mesh fragments that surround the organic forms in the training data do not appear in any output. Such detail loss and surface smoothing were already visible in the training samples (Fig. 2), so they cannot be attributed to the input conditions alone.

4.3. Cost

Throughput showed no systematic dependence on either the input condition or the adapter. All sixteen runs of Phases 1 and 2 fell between 7.5 and 8.0 fps (Table 2). Memory never became the constraint — peak allocation was 4439 MiB, leaving about 3.6 GB of the 8 GB unused. The mean card temperatures were 86 to 87 °C, with brief clock reductions to 1650 MHz (Table 2).

Removing the second denoising step raised throughput from 7.5 to 11.8 fps, while the Style Index stayed at 14.

4.4. Generating the Input on the Same GPU

In Phase 4, generating the C07 input live in TouchDesigner instead of supplying it as a still frame reduced throughput from 7.8 to 6.6 fps, a loss of about 15 %, while memory and GPU utilisation were unchanged (Table 2). The mean clock speed was 6 % lower in the live run, which may itself follow from the added load. The run was not rated visually, since its input is a different frame.

	fps	VRAM MiB	Load %	Clock MHz	Temp °C
P1, adapter	7.5–8.0	4174–4379	90–92	2161–2228	89–94
P2, no adapter	7.5–8.0	4159–4312	90–92	2183–2231	88–92
C01, two steps	7.5	4379	91	2228	92
C01, one step	11.8	4251	91	2284	89
C07, still frame	7.8	4235	92	2192	89
C07, live input	6.6	4241	92	2053	90

Table 2: *Throughput and GPU load. Values are means, temperature the maximum per run. The top block gives ranges across the eight runs of Phases 1 (P1) and 2 (P2), the rows below are single runs.*

5. Discussion and Conclusion

5.1. What Determines Whether the Style Appears

The cascade did not predict the result, and the settings cannot explain it. Every condition ran at the same noise level before denoising, which fixes how far the output can move away from the input [MHS*22]. What may differ is how much of each input is left once that noise is added. Contrast does not order the conditions, as Section 4.1 shows. What the stronger ones do share is a figure that stands apart from its background and a composition close to the training renders, in three of four cases with colour.

The same coupling between local brightness and placement is reported by Akten et al. in a conditional GAN, and a comparable pattern appears here in a few-step diffusion pipeline [AFG19].

The trigger token, by contrast, proved ineffective. When the adapter remained active but *orgflw* was omitted, control samples stayed nearly identical to outputs containing the token at every checkpoint (Fig. 2). Because all 150 captions were identical, the trigger never occurred separately from the subject description, leaving the model with no basis for distinguishing between the two. Koh et al. describe this form of entanglement [KMK*26].

5.2. What Personalisation Costs

The adapter's weights are merged before streaming, so the model that runs is the same size as the base model and performs the same operations, which is why it costs nothing measurable [HSW*22]. However, throughput depended on the denoising schedule and on the image source sharing the same GPU. For the motivating use case, the remaining GPU budget determines how complex that source can be.

Throughput is only half of the cost. Where the input is rendered in advance rather than generated live, it costs render time instead of GPU share. Since the pipeline runs at the same speed whatever reduction of the original it is given, what the input costs to produce decides whether a condition is usable for this workflow. I therefore rendered the same frame in Blender once in each mode, at one consistent Cycles setting that is not the production setting quoted in Section 1.1. The viewport solid, flat colour and binary variants appear without perceptible delay and Blender's interface shows no per-frame time for them. Eevee takes 0.8 s. Cycles takes 57.2 s with the low-poly geometry and 60.3 s with the full mesh. Reducing the

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polygon count therefore saves about 5 % of render time and does not change the output, while leaving out light transport cuts it from about a minute to under a second.

5.3. Limitations

This study covers one artist, one style, one dataset of 150 images, and one GPU. All outputs were rated by me, the author, from a single frame and seed per condition. Because the Style Index sums four equally weighted five-point criteria, it ranks conditions rather than measuring distance between them, and rating repeatability was not tested. Notably, the cascade is a designed ordering, not a computed measure of distance to the training distribution. I developed the figure-ground and brightness explanations after rating, from the same eight cases they describe, so they fit these results by construction. Neither is measured, and testing them would need new images. Style was assessed only on still frames, which provides no information about moving input or temporal coherence. The low surface detail already appears in the training samples, preventing this study from separating training effects from pipeline effects. A single run of a single network was used for the live-input measurement. Without TensorRT, every throughput figure reported here is a lower bound. Of the two main results, the cost finding should transfer more readily, since it follows from adapter merging rather than from what the adapter learned. The brightness finding transfers only in part. Akten et al. report the same coupling [AFG19], but in my case the model had little dark content to place, because all 150 training images had bright backgrounds. Only a training set with mixed backgrounds could show whether the data causes this gap.

5.4. Conclusion

Returning to the question that motivated this work, on this hardware the input determined how strongly the style appeared, while the adapter added no measurable cost. The Style Index spanned eight points across the eight conditions with every other setting fixed, yet the designed cascade did not predict that spread. Which input properties produce it remains open, since the eight conditions differ in colour, contrast, brightness and form at once and this study cannot separate them. Polygon count left the output unchanged and cut render cost by only about 5 %. A flat colour reduction that costs no appreciable render time landed two points below a path-traced render on the 4–20 scale, so within this pipeline it is a usable substitute for the expensive input. Cost-wise, the fused adapter left throughput and memory unchanged across sixteen runs; one instead of two denoising steps raised throughput about 1.57-fold at an unchanged Style Index, while generating the input on the same GPU cut throughput by roughly 15 %.

The pipeline itself is not yet a substitute for the offline render. It stays under 12 fps, well below the 24 fps of conventional animation and too slow for me to use live. What it produces is a related but different look. No condition scored above 14 of 20, so the remaining distance to my own style lies in the pipeline and the training rather than in the choice of input.

5.5. Future Work

Future work can test all hypotheses more directly, though some are easier to isolate than others. Figure-ground separation would need a defined measure first, since the contrast measure I computed does not capture it. Colour and contrast can be varied directly. Keeping geometry and luminance fixed and changing only the colours would show whether the input needs colour at all, and if so, whether it needs the colours of the training images in particular. Contrast can be varied in two ways. Amplifying the light and dark areas of the viewport solid condition changes how far image regions differ, and blurring the edges of the binary mask changes how sharply they meet.

A second evaluator and blind re-rating would show how far the ratings depend on individual judgment. Moving input should also be tested, either as pre-rendered flat colour animations switched during performance or as low-poly geometry processed live in TouchDesigner and reduced to a mask. Retraining with varied captions and backgrounds could make the trigger token usable as a control. Repeating the series on a release with production TensorRT support for this GPU generation would show the available headroom [Hin25]. Applying the same protocol to another style would show how far these findings carry beyond this single case.

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